Amirhosein Mohaddesi

AI Systems Architect | Data-Centric Autonomy | Agentic Systems | ROS2 Multi-Robot Infrastructure

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Summary

Ph.D. AI Systems Architect specializing in machine learning infrastructure for autonomous multi-robot teams. I build production-grade ROS2 platforms with high-throughput C++/Python telemetry pipelines, PyTorch training workflows, and policy-aware agentic orchestration via LangGraph and LangChain where language guidance measurably reduces coordination overhead. Authorized to work in the U.S. without sponsorship.

Research Experience

Graduate Researcher — **CA, USA**
CARL Lab, University of California, Irvine — *Sep 2021 – Jun 2025*

ROS2 Multi-Robot Navigation Platform, GitHub

- *Problem:* Webots-only prototypes could not scale swarm size, sensor fidelity, or fleet telemetry needed to train navigation policies from human trajectory data.
- *Solution:* Architected a ROS2 Humble simulation platform with SLAM Toolbox, Nav2, map_merge, frontier exploration, and C++/Python nodes for navigation, mapping, and high-rate state logging into PyTorch-ready datasets.
- *Outcome:* Enabled larger multi-agent experiments with RViz live telemetry and quantified task efficiency, coverage, and cooperation across Route, Survey, and Mixed navigation strategies.
- *Learning:* Reliable autonomy infrastructure must precede learning—telemetry fidelity and modular ROS2 graph design determine whether imitation learning and agentic planners can ship to sim-to-real.

Communication-Aware Multi-Robot Coordination

- *Problem:* Homogeneous navigation policies caused redundant exploration and coordination failures as team size grew.
- *Solution:* Designed a communication-aware coordination layer for goal sharing and path deconfliction; integrated agentic planning hooks via LangGraph and LangChain for language-guided fleet decisions when classical policies stalled.
- *Outcome:* Mixed human-inspired strategies delivered the strongest balance of exploration breadth and task completion time across disaster-response simulation scenarios.
- *Learning:* Strategy diversity plus explicit inter-robot messaging beats monolithic policies for scalable team autonomy.

Strategy Diversity in Robot Teams, GitHub

- *Problem:* No empirical baseline existed for how heterogeneous navigation behaviors affect team-level efficiency in embodied multi-agent settings.
- *Solution:* Built a Webots multi-agent testbed with Clearpath PR2 robots and a C++ obstacle-avoidance controller; ran controlled sweeps over Route, Survey, and Mixed strategy assignments.
- *Outcome:* Published findings on strategy diversity at SAB 2024; demonstrated measurable gains in coverage and coordination effectiveness for exploration-style missions.
- *Learning:* Heterogeneity is a controllable systems knob—treat strategy assignment as infrastructure, not an afterthought.

Autonomous Telepresence Navigation

- *Problem:* Manual telepresence navigation imposed high cognitive load and degraded task performance during remote scavenger-hunt operations.
- *Solution:* Shipped real-time SLAM navigation in ROS with a PyQt operator GUI and instrumentation for cognitive load, spatial awareness, and movement efficiency.
- *Outcome:* Autonomous navigation reduced operator burden and improved usability; results presented at IEEE ICDL 2024.
- *Learning:* Autonomy wins when telemetry-backed UX metrics prove reduced operator cost, not when autonomy is added for its own sake.

Research Assistant — **CA, USA**
NMI Lab, University of California, Irvine — *Jul 2020 – Jul 2021*

Embedded ML Quantization

- *Problem:* Full-precision spiking neural networks exceeded power budgets on embedded inference targets.
- *Solution:* Implemented 8-bit quantized SNNs in PyTorch with a custom quantization scheme for deployment-constrained hardware.
- *Outcome:* Cut energy use 12%–18% on MNIST/CIFAR benchmarks with only 3%–7% accuracy loss.
- *Learning:* Quantization trade-offs are systems decisions—profile accuracy, power, and latency jointly before picking a bit width.

Selected Projects

Policy-Aware Agentic Task Execution

2025

- *Problem:* Multi-robot missions needed a governed execution layer that could translate high-level objectives into safe, auditable Nav2 actions without bypassing fleet policy constraints.
- *Solution:* Built a LangGraph and LangChain mission-execution orchestrator over ROS2/Nav2 with schema-validated tool calls, policy gates, and telemetry feedback loops for runtime state.
- *Outcome:* Delivered language-guided task execution that respects navigation boundaries while coordinating multi-robot goals in simulation.
- *Learning:* Agentic autonomy over real robot stacks requires explicit policy enforcement at every tool boundary—orchestration without guardrails is not production-ready.

Career Conversation Agent, Hugging Face Demo

2025

- *Problem:* Static portfolios could not answer nuanced, context-dependent questions from recruiters and collaborators.
- *Solution:* Deployed a persona-grounded conversational agent in Python and Gradio with embedding-backed context retrieval and public Hugging Face Space integration.
- *Outcome:* Live demo embedded on personal site; enables interactive career Q&A without manual follow-up.
- *Learning:* Production agent UX requires tight retrieval boundaries—ground answers in verified profile data, not open-ended generation.

ROS2 Distributed Multi-Robot Control Stack, CARL Lab GitHub

2024

- *Problem:* Single-process simulators could not stress-test distributed mapping and frontier coordination under realistic ROS2 timing.
- *Solution:* Integrated SLAM Toolbox, Nav2, map_merge, and frontier exploration into a ROS2 Humble disaster-response scenario with DDP-ready PyTorch training hooks.
- *Outcome:* Delivered a reusable research platform for collaborative navigation, mapping, and fleet telemetry capture.
- *Learning:* Distributed ROS2 graphs expose race conditions early—design telemetry and merge policies before scaling agent count.

Publications

- Mohaddesi, S.A.; Hegarty, M.; Chrastil, E.R.; Krichmar, J.L. Benefit of Varying Navigation Strategies in Robot Teams. Proceedings of SAB 2024, Lecture Notes in Computer Science, Vol. 14993, pp. 63–77, Springer, 2024.
- Pan, G.; Weiss, T.; Mohaddesi, S.A.; Szura, J.W.; Krichmar, J.L. Navigation and Cognitive Load in Telepresence Robots. IEEE ICDL 2024, pp. 1–6, 2024.

Skills

- **Agentic Systems**: LangGraph, LangChain, RAG, tool-use orchestration, policy validation, prompt engineering
- **Autonomy & Robotics**: ROS2 Humble, Nav2, SLAM Toolbox, Gazebo, Webots, sim-to-real, multi-robot coordination
- **ML Infrastructure**: PyTorch, distributed training (DDP), CNN/RNN/Transformers, RL (DQN, PPO), Hugging Face
- **Systems Programming**: Python, C++, Bash; high-throughput telemetry/logging pipelines; Linux, Docker, Git

Education

Irvine, CA

University of California, Irvine *Ph.D. in Information and Computer Science* *Sep 2019 – Jun 2025*

- GPA: 3.93/4.0 **Focus:** Robotics, ML infrastructure, multi-agent systems
- Dean's Award, Donald Bren School of ICS

Tehran, Iran

Sharif University of Technology *B.S. in Computer Engineering* *Sep 2015 – Jun 2019*

- GPA: 3.95/4.0

Achievements

- **Direct PhD fellowship** – Donald Bren School of ICS **UCI, Irvine, CA** *2019*
- **Dean's Award** – Donald Bren School of ICS **UCI, Irvine, CA** *2019*
- **Silver Medal** – Iran's National Physics Olympiad **Tehran, Iran** *2013*